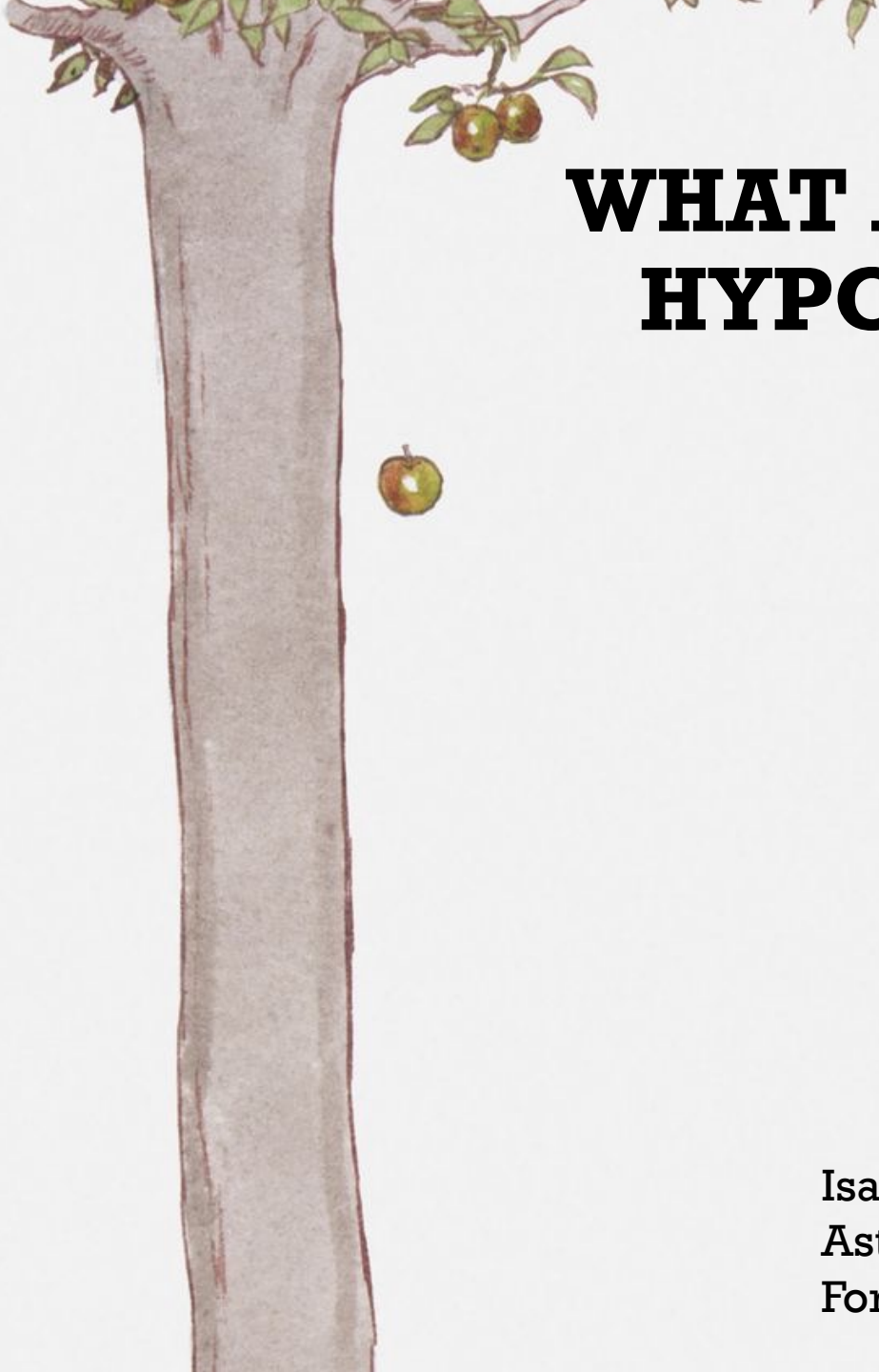


An illustration of a tree trunk on the left side of the slide. A single apple is shown in mid-air, having just fallen from a branch. The branch has several green leaves and a few more apples.

WHAT ARE THEORIES, HYPOTHESES, AND LAWS?

Class 2

Isaac Newton (1643-1727)
Astronomer, mathematician, physicist
Formulated the Law of Gravity



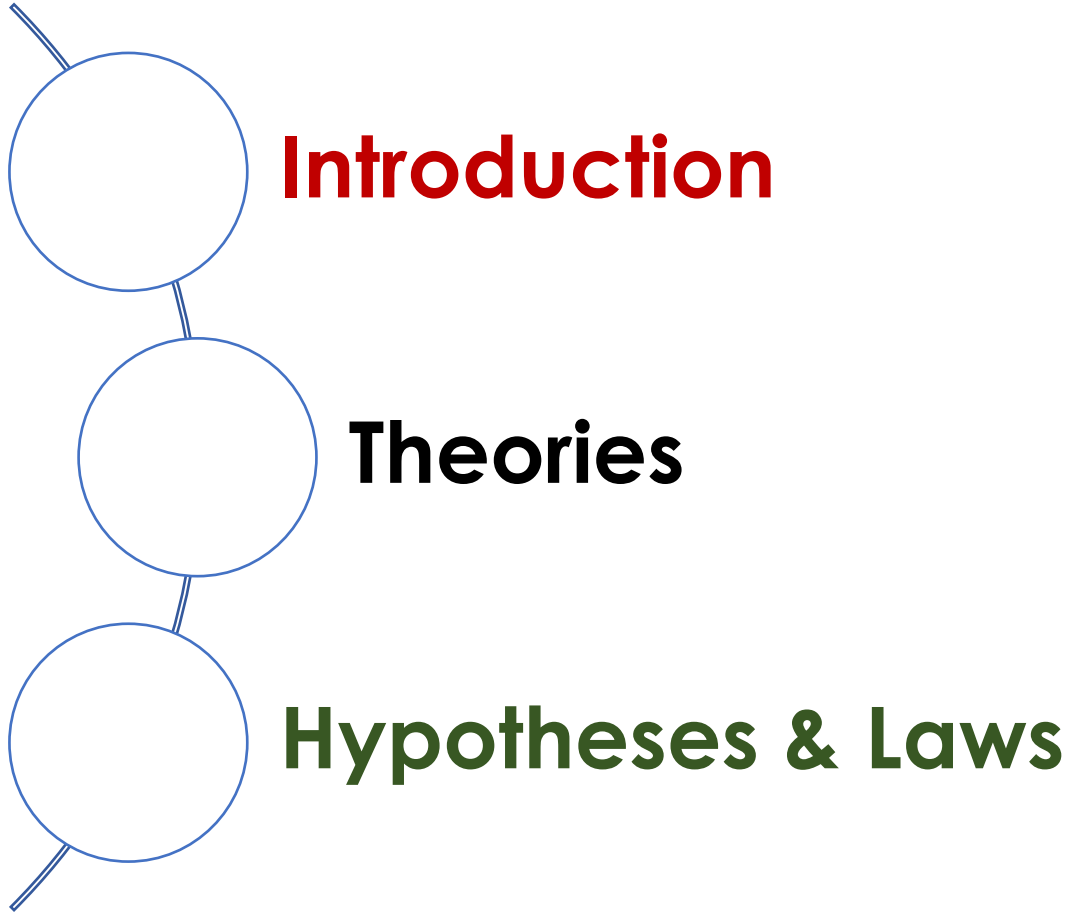
ANNOUNCEMENTS

Please sign up to your group on Canvas (in case you have not done it)

‘Canvas’ → ‘People’ → ‘In-class Activities’ (you can drag and drop your name into your group)

Please, provide me with your group name

The Plan!



INTRODUCTION

- Is truth the aim of science?

Of course, right?

INTRODUCTION

- Most people believe that science is a *reliable* way to acquire knowledge (justified true beliefs)
 - objective
 - free of prejudice or unbiased
 - free of superstition
 - trustworthy
 - ...
- Something is done scientifically = something is true!?

INTRODUCTION

- Most people believe that science is a reliable way to acquire knowledge (justified true beliefs)
- But why should we believe what science says about the world?
 - This is a question of *justification of scientific knowledge*
 - This question asks not so much what science claims about the world but more how science **proves** what it does claim and why it **gives us reason** to believe that these claims are true.

INTRODUCTION

- Is the success of scientific theories a good indicator of their truth?



Reflect and discuss:

1. *“Science gains credibility from its phenomenal success. Airplanes fly. Smallpox has been done away with. Humble uranium gets turned into electricity, which in turn keeps the Bud Light cold. This record of success may be enough to give us responsible warrant to believe that science, by and large, gets at the truth about the world ...”*

Can the phenomenal success of science be considered sufficient in the justification of scientific claims? Why?

INTRODUCTION

- Science is largely about *unobservable* (theoretical) entities (“electrons”, “black holes”, etc).
 - For instance, the sentence “iron turns red when it is heated” can be verified because we can directly observe the iron, the redness, the heat, etc. The phenomena are *observable*.
 - But how can we verify the sentence “Helium atoms contain two electrons”?
 - We cannot directly observe these things!

INTRODUCTION

When we “reduce” electron to its experiential consequences, we get something like bubbles in a cloud chamber. We don’t see electrons directly. We see the bubble trail it leaves.



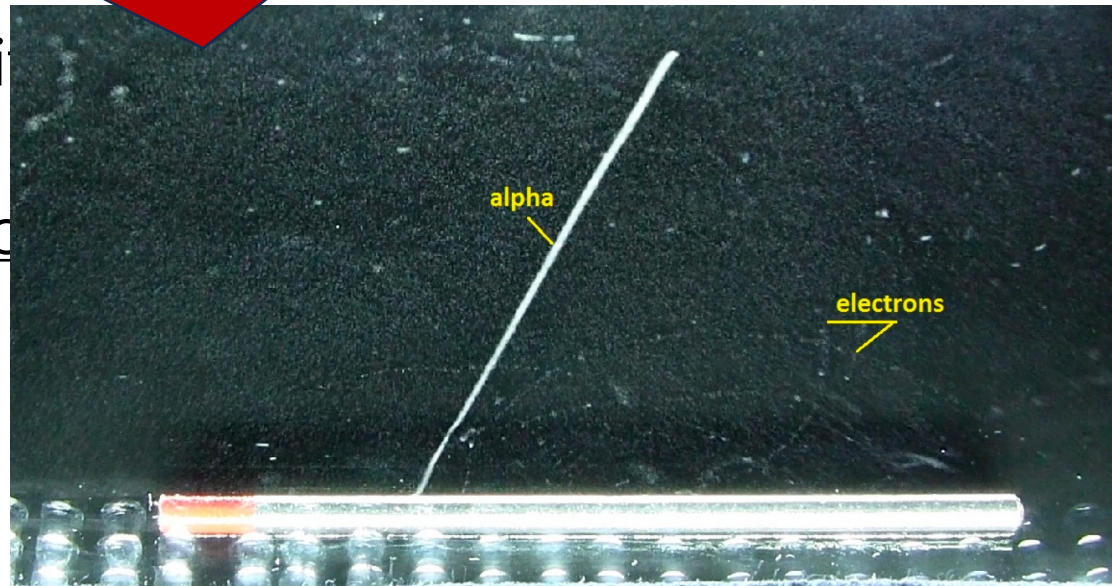
Cloud Chamber

heat, etc. The phenomena is observable.

- But how can we verify

directly c

ain two



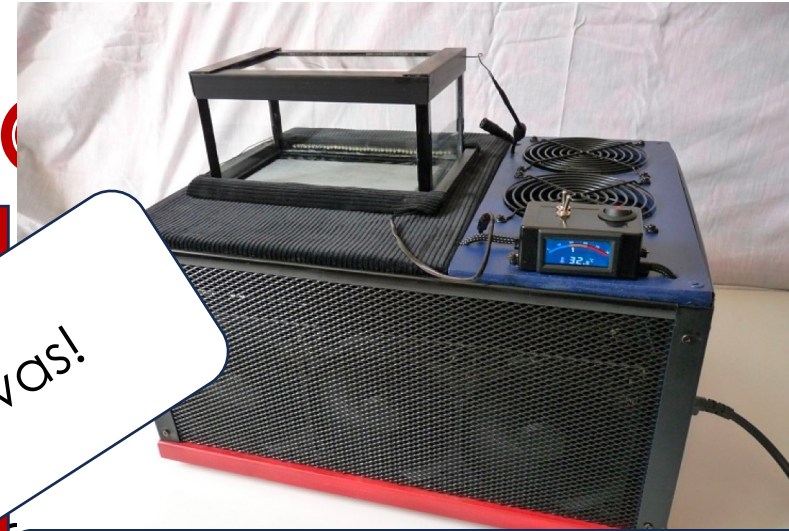
INTRODUCTION

When we “reduce” electron to its experiential consequences, we get something like bubbles in a cloud chamber. We don't see electrons directly. We see the bubbles they leave.

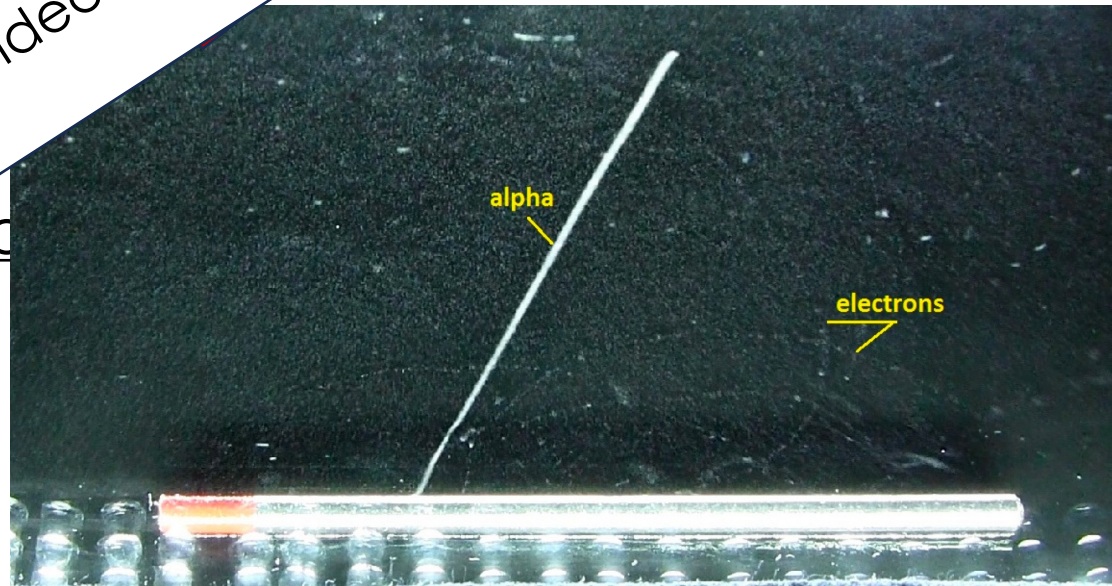
heat, etc. The phenomena is observable.

- But how can

I posted the videos about this on Canvas!



Cloud Chamber



gain two

INTRODUCTION

What counts as success of a theory that is about things that can never be directly observed or experienced?

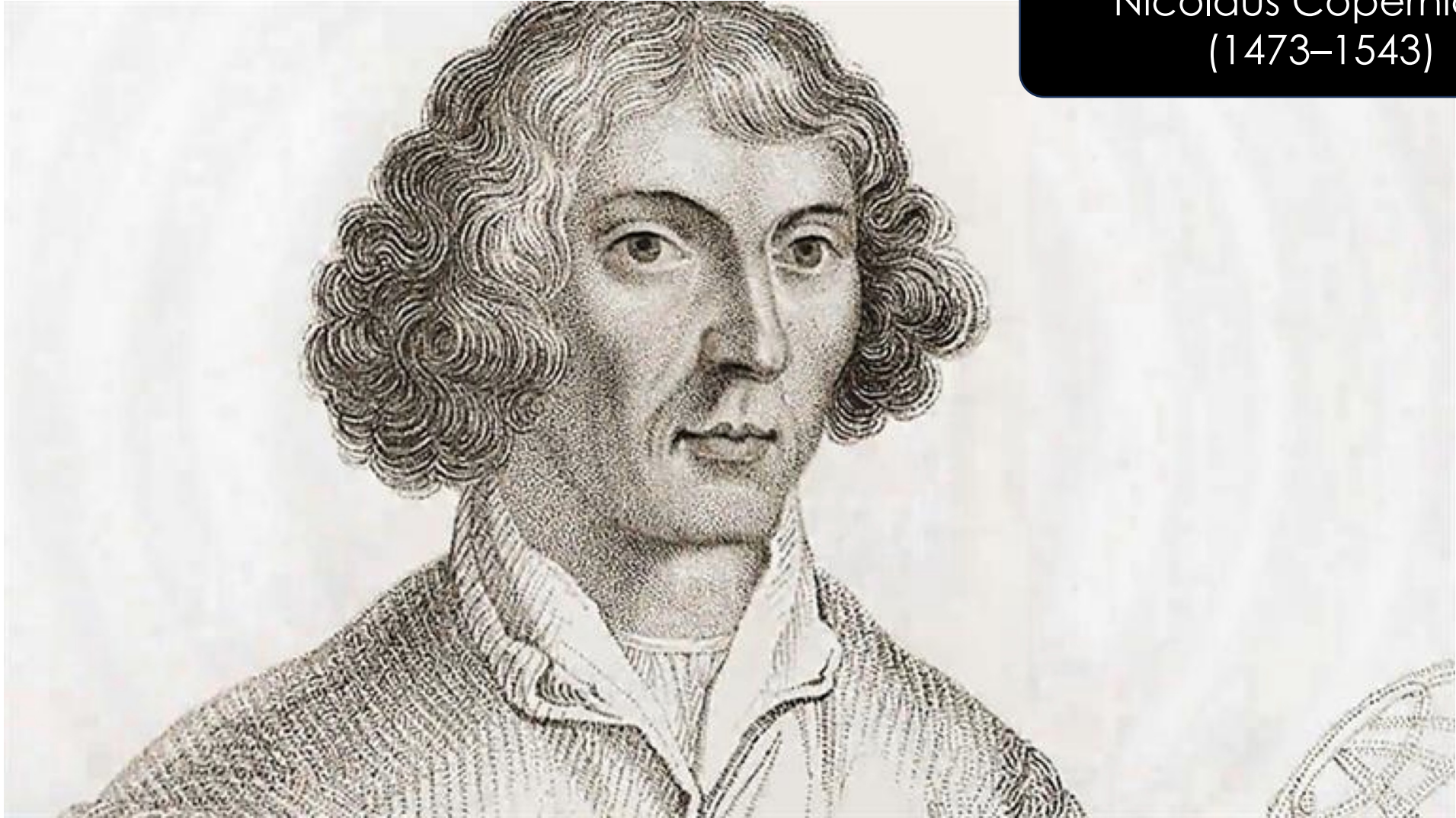
- **(1)** The criteria and measures of success are **proclaimed** by scientists themselves. Science, in this sense, is a largely **self-regulating** business.

INTRODUCTION

(2) Scientific theories change over time.

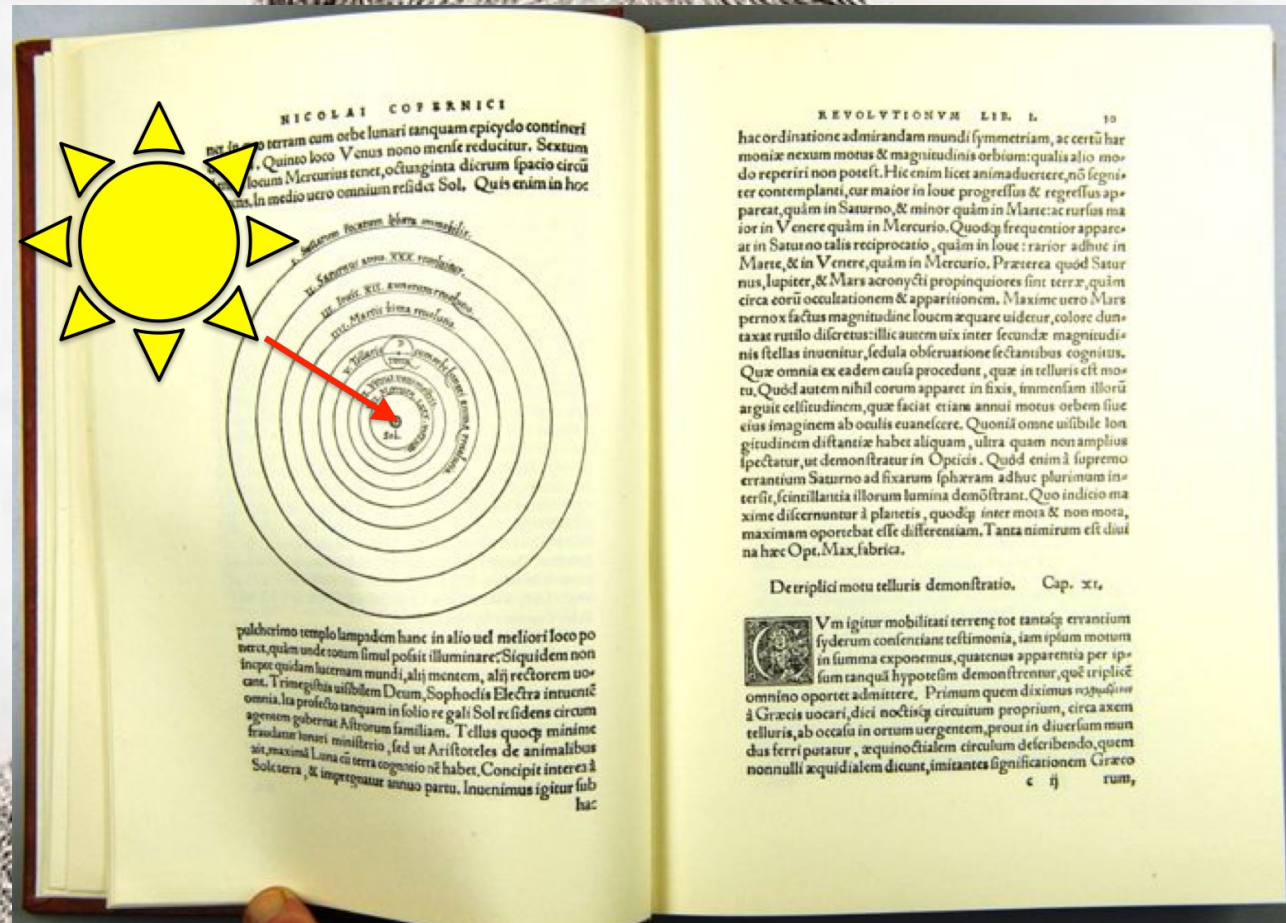
INTRODUCTION

Nicolaus Copernicus
(1473–1543)



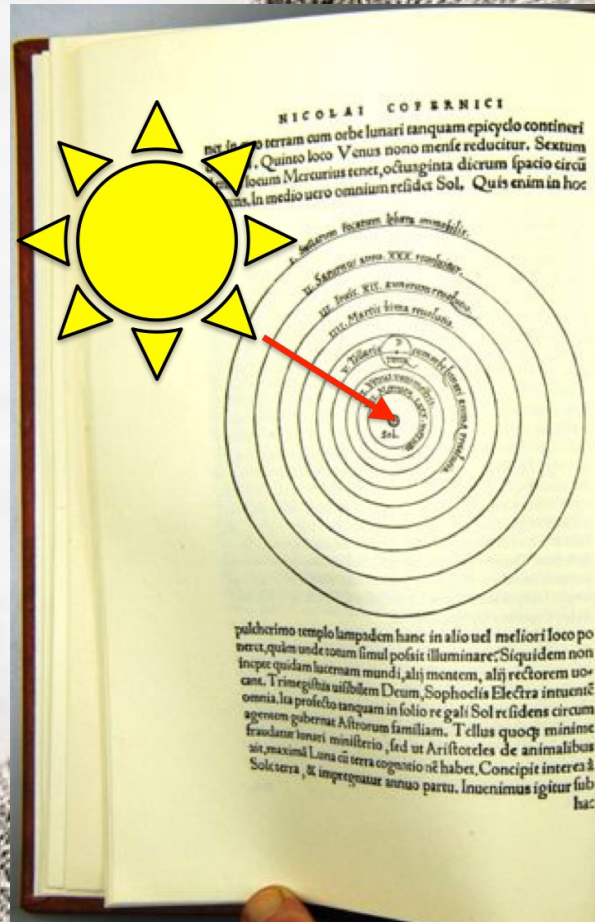
INTRODUCTION

Nicolaus Copernicus
(1473–1543)



INTRODUCTION

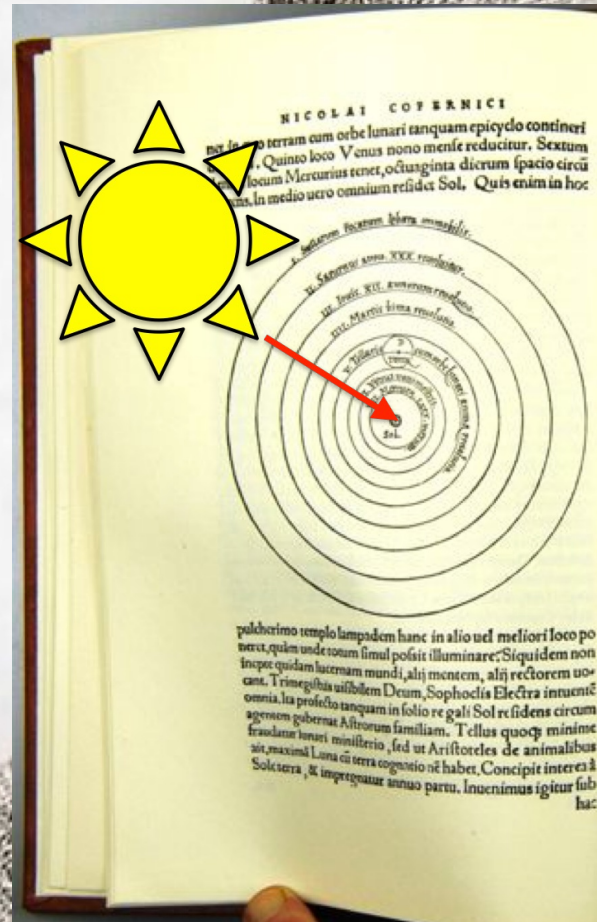
Nicolaus Copernicus
(1473–1543)



“Since [the astronomer] cannot in any way attain to the true causes, he will **adopt whatever suppositions enable the motions to be computed correctly** [...] For these hypotheses **need not be true nor even probable**. On the contrary, if they provide a calculus consistent with the observations, that alone is enough [...] they are not put forward to convince anyone that they are true, but **merely to provide a reliable basis for computation**.”

INTRODUCTION

Nicolaus Copernicus
(1473–1543)



“Since [the astronomer] cannot in any way attain to the true causes, he will **adopt whatever suppositions enable the motions to be computed correctly** [...] For these hypotheses **need not be true nor even probable**. On the contrary, if they provide a calculus consistent with the observations, that alone is enough [...] they are not put forward to convince anyone that they are true, but **merely to provide a reliable basis for computation**.”

Science aims for *empirical adequacy* only, not truth?

INTRODUCTION

Ah! But the success of today's theories must be good evidence of their truth?!?

INTRODUCTION

Ah! But the success of today's theories must be good evidence of their truth?!?

Consider this:

Many, many past theories were successful in their day.

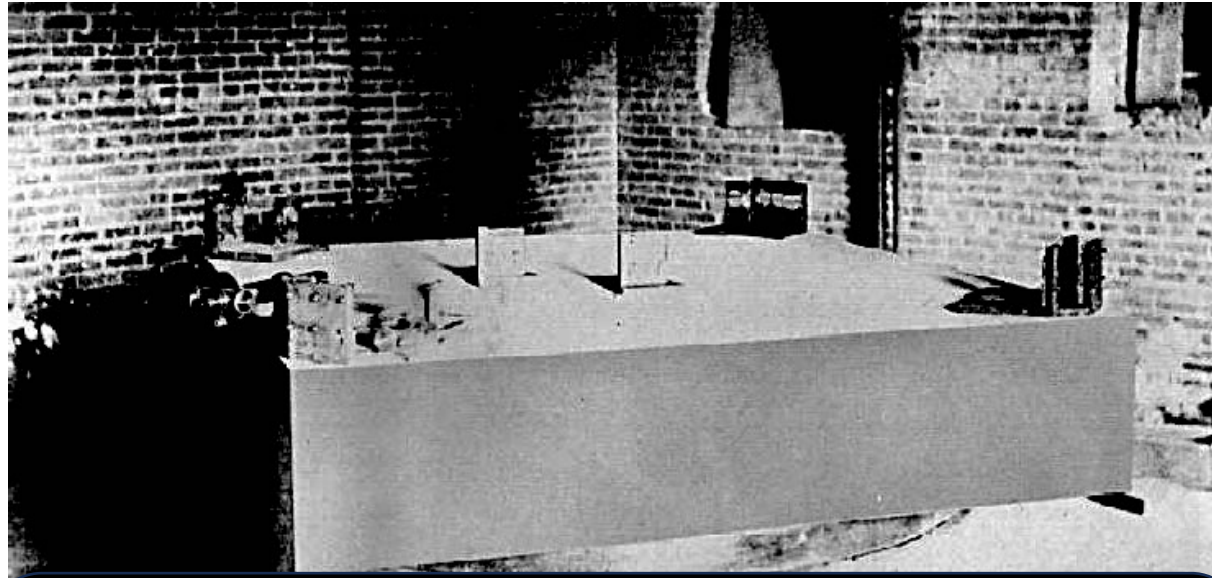
Most have turned out to be false.

Therefore, by induction, success is not a good indication of truth.
In fact, we should *infer from evidence* that current theories will likely be false!

(Called: the pessimistic induction)

INTRODUCTION

(3) To say that something is done scientifically is not to say simply that it is done with success.



The Michelson-Morley experiment (1887) is a famous experiment in physics that tried and failed to detect the motion of the Earth through the 'aether'.

INTRODUCTION

(3) To say that something is done scientifically is not to say simply that it is done with success.

- Success may be the result, but it is not the methodological structure.

INTRODUCTION

- What is this methodological structure (or scientific method), and what about it gives us reason to think it produces truths?

WHAT IS SCIENCE?



- The scientific method cannot be identified with any particular set of experimental or observational procedures.
 - Methods vary:
 - In physics, biology, and chemistry, controlled experiments to test hypotheses are often possible.
 - Epidemiology, medicine, ecology often employ “natural” experiments
 - In astronomy and geology, hypotheses must usually be tested through observation.

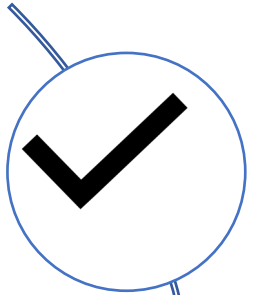
WHAT IS SCIENCE?

Last time!

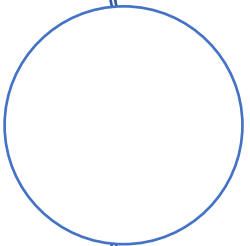
- The scientific method cannot be identified with a particular set of experimental or observational procedures
 - Methods vary:
 - In physics, biology, and chemistry, controlled experiments to test hypotheses are often possible
 - Epidemiology, medicine, and social sciences often employ “natural” experiments
 - In astronomy and geology, hypotheses must usually be tested through observation

There are nonetheless some common features that are significant to the justification of scientific claims

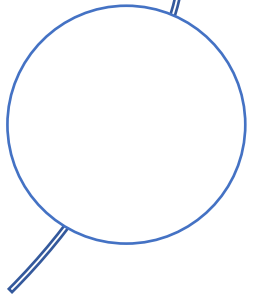
The Plan!



Introduction

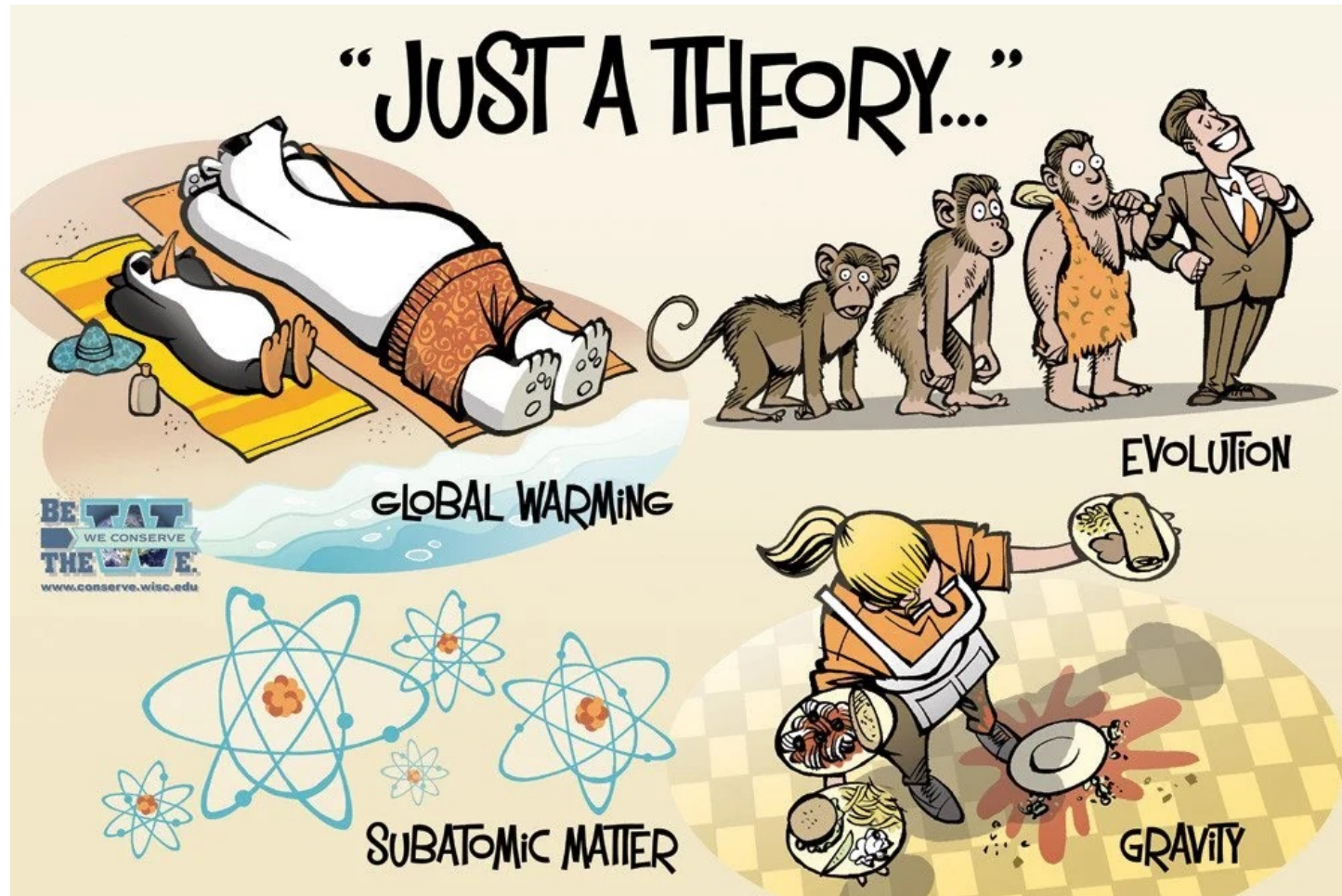


Theories



Hypotheses & Laws

THEORIES



THEORIES



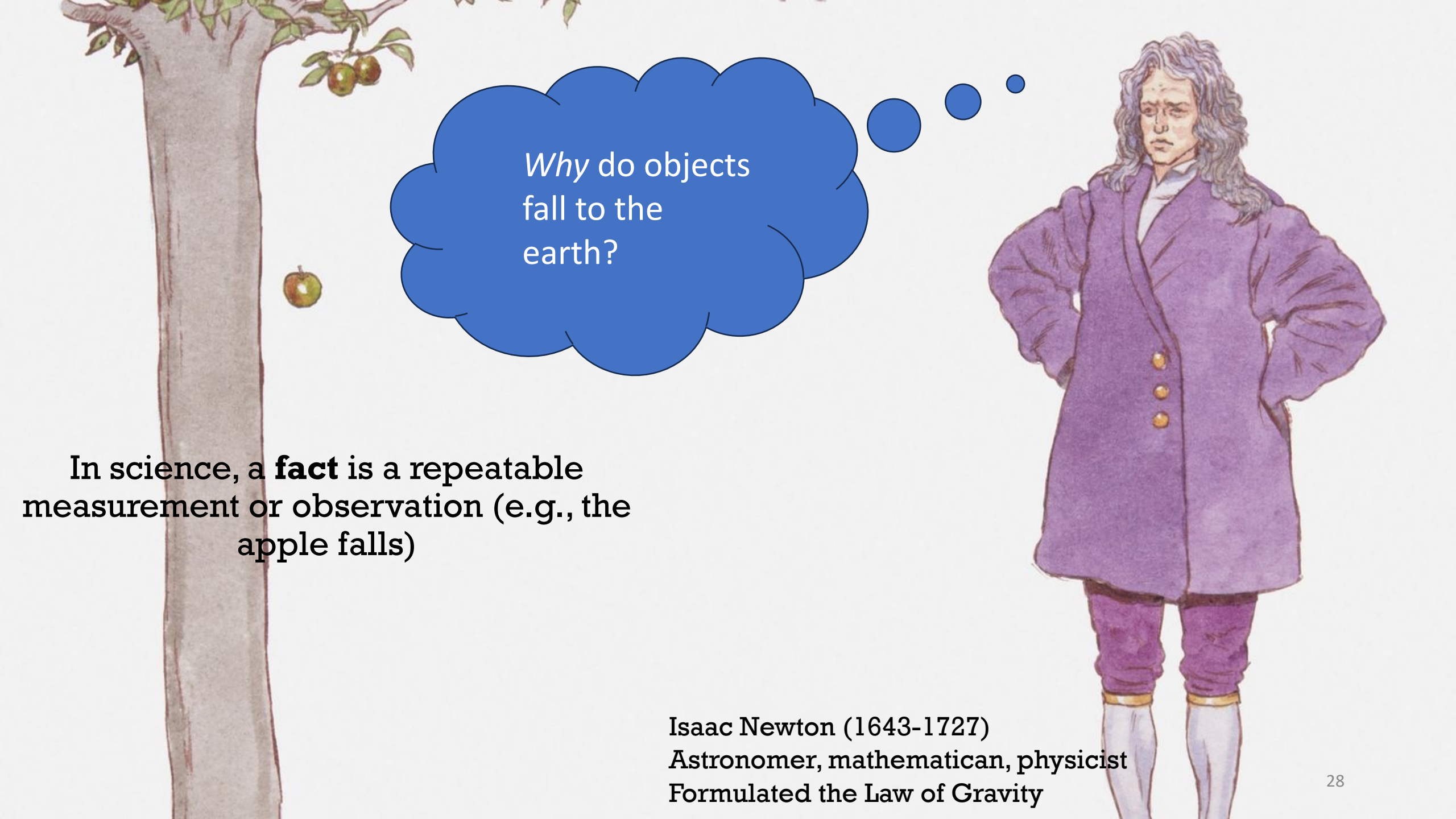
Reflect and discuss:

2. *“When you stop talking about theories, you stop talking about science.”*

- **Explain**

THEORIES

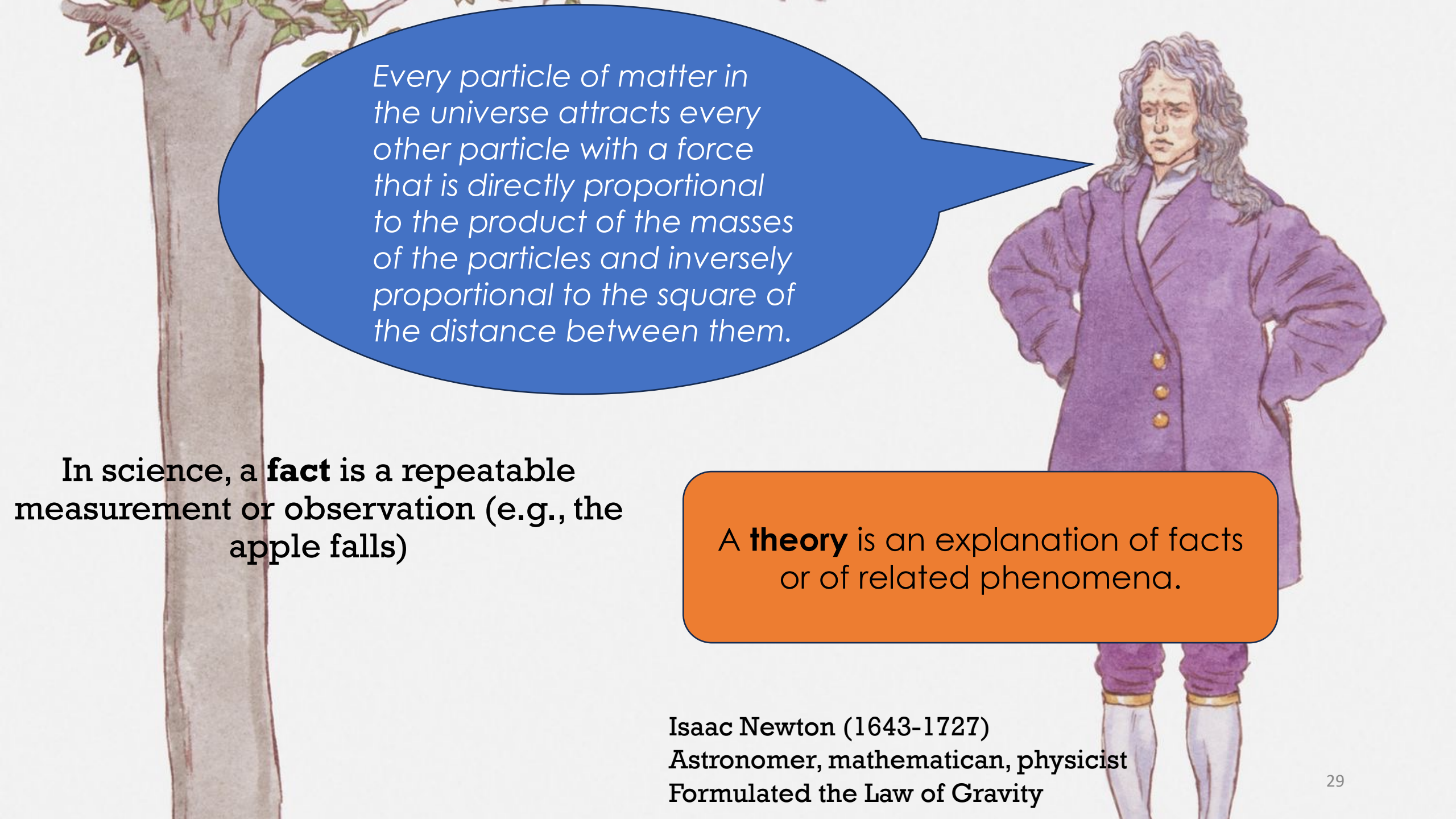
- The formulation of *theories* is the primary product of the scientific enterprise
 - **Theory (Def.)**: Any claim about the external world, general or specific, well justified or speculative, is a theory. Theories are the elements of expansive knowledge in that they always make claims to more than is told by the evidence.

An illustration of Isaac Newton standing next to a tree. A thought bubble from his head contains the text 'Why do objects fall to the earth?'. An apple is shown falling from the tree. Newton is wearing a long purple coat and has his hands on his hips.

Why do objects
fall to the
earth?

In science, a **fact** is a repeatable
measurement or observation (e.g., the
apple falls)

Isaac Newton (1643-1727)
Astronomer, mathematician, physicist
Formulated the Law of Gravity



Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them.

In science, a **fact** is a repeatable measurement or observation (e.g., the apple falls)

A **theory** is an explanation of facts or of related phenomena.

Isaac Newton (1643-1727)
Astronomer, mathematician, physicist
Formulated the Law of Gravity

THEORIES

- *The aim of science:*
 - To come up with theories that help us explain what we directly observe (that help us make sense of our direct observations of the world)

THEORIES



Reflect and discuss:

3. “Science is not after simply a careful, methodological account of the observable world.”

- **Explain**

THEORIES

- Science aims
 - to understand the composition and underlying causes of the ***manifest*** phenomena and thereby make sense of the phenomenal world we inhabit.

THEORIES

- Science aims
 - to understand the composition and underlying causes of the ***manifest*** phenomena and thereby make sense of the phenomenal world we inhabit.



What does
manifest mean?

THEORIES

- Science aims
 - to understand the composition and underlying causes of the **manifest** phenomena and thereby make sense of the phenomenal world we inhabit.

observable; events and
occurrences
that can be perceived by
the senses and experienced

THEORIES

- Science aims
 - to understand the composition and underlying causes of the **manifest** phenomena and thereby make sense of the phenomenal world we inhabit.
 - to offer **theories** that explain why the phenomena occur and what causes them.
- Science
 - is 'speaking about things **beyond** the evidence'
 - brings on a degree of uncertainty
- Theories are the primary subjects of **justification**

THEORIES



Reflect and discuss:

4. “One sort of theory will be of claims that are only about the unobservable world ... All claims of this sort describe solely what the describers cannot experience. The other type of theoretical claim is needed to break the isolation.”

What two kinds of claims are being considered here? Describe with examples.

THEORIES

- 1) Claims that are only about unobservables without any direct impact on the phenomenal world
 - All bayrons (protons) are made of quark.
 - Helium atoms contain two electrons
- 2) Claims about the interactions between unobservables and observable.
 - Cloud Chamber

INTRODUCTION

When we “reduce” electron to its experiential consequences, we get something like bubbles in a cloud chamber. We don’t see electrons directly. We see the bubble trail it leaves.



Cloud Chamber

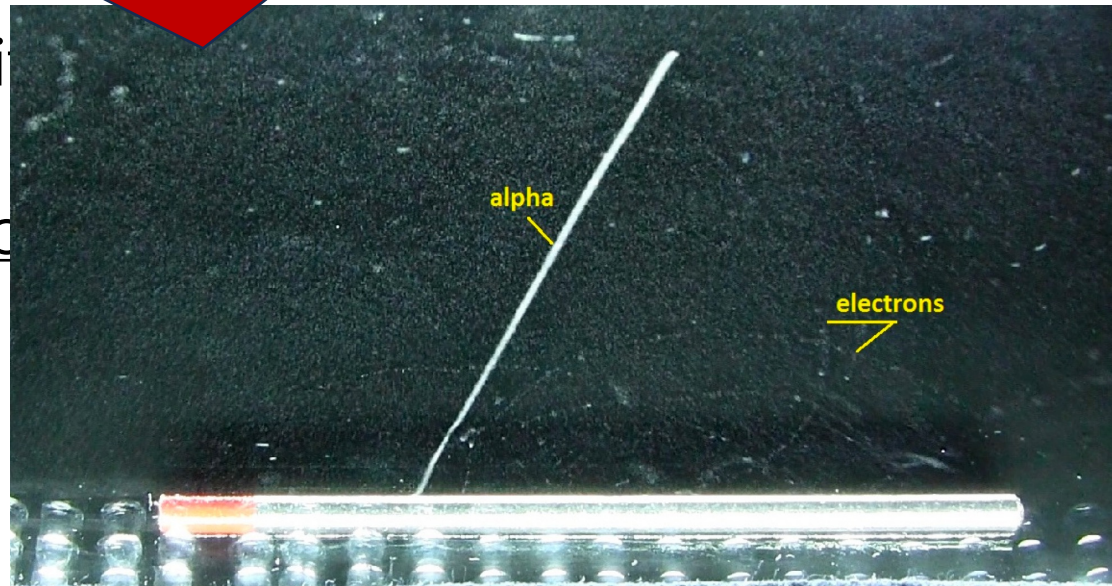
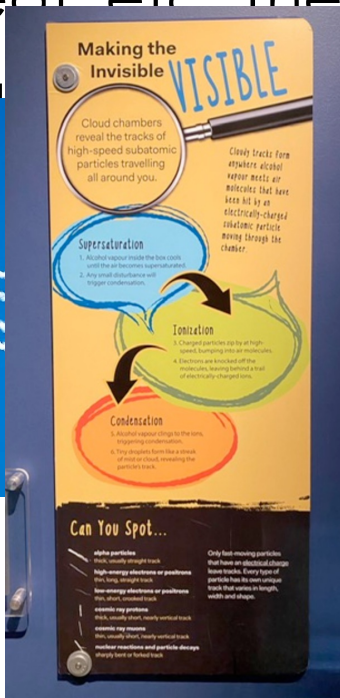
heat etc. The phenomena is observable.

• But

we verify

gain two

directly c



THEORIES

- 1) Claims that are only about unobservables without any direct impact on the phenomenal world
 - All bayrons (protons) are made of quark.
 - Helium atoms contain two electrons
- 2) Claims about the interactions between unobservables and observable.
 - Cloud Chamber



What have both claims in common?

THEORIES

- 1) Claims that are only about unobservables without any direct impact on the phenomenal world
 - All bayrons (protons) are made of quark.
 - Helium atoms contain two electrons
- 2) Claims about the interactions between unobservables and observable.
 - Cloud Chamber



Both claims describe more than what is apparent, and in doing so they draw the question of justification.

THEORIES

1) Claims that are only about unobservables without any direct impact on the phenomenal world

- All bayrons (protons) are made of
- Helium atoms contain two electro

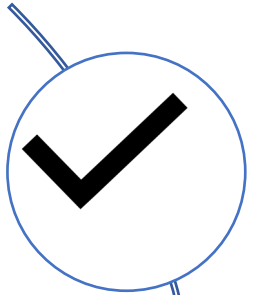
When is a theory to be believed?
To what degree is it to be believed?

2) Claims about the interactions between

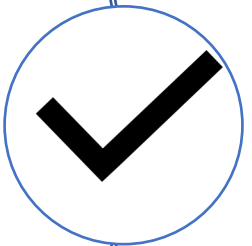
- Cloud Chamber

Both claims describe more than what is apparent, and in doing so they draw the question of justification.

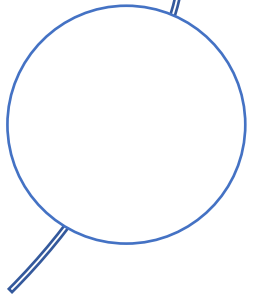
The Plan!



Introduction



Theories



Hypotheses & Laws

HYPOTHESES & LAWS



Reflect and discuss:

5. What differentiates the following terms: theory, hypothesis, law?

THEORIES



- The formulation of *theories* is the primary product of the scientific enterprise
 - **Theory (Def.)**: Any claim about the external world, general or specific, well justified or speculative, is a theory. Theories are the elements of expansive knowledge in that they always make claims to more than is told by the evidence.

HYPOTHESES & LAWS

- **Hypothesis (Def.):**
 - The particular theory that is being tested in a certain case is the hypothesis in that case.
 - This term is used to identify a theory (or theories) that is in need of testing.
 - Thus a hypothetical claim is made without much commitment to its truth. It is a suggestion in want of confirmation.

HYPOTHESES & LAWS



- Scientists aim to verify their hypotheses by appealing to observations they make under special circumstances (e.g., experiments)
- Scientists believe (“accept as true enough to believe”) those hypotheses that are *highly confirmed* (those that have been verified by observation).

HYPOTHESES & LAWS



- A simple example:
 - We start with a *hypothesis*: “High air pressure is associated with fair, stable weather.”

HYPOTHESES & LAWS



- A simple example:
 - We start with a *hypothesis*: “High air pressure is associated with fair, stable weather.”
 - We make *predictions* (these are implications that follow from the truth of the hypothesis)
 - We *will* find high air pressure on days when the weather is fair and stable

HYPOTHESES & LAWS



- A simple example:
 - We start with a *hypothesis*: “High air pressure is associated with fair, stable weather.”
 - We make *predictions* (these are implications that follow from the truth of the hypothesis)
 - We *will* find high air pressure on days when the weather is fair and stable
 - We go out and *test* this
 - Go out on a fair and stable day, check whether we observe high air pressure

HYPOTHESES & LAWS



- A simple example:

- We start with a *hypothesis*: “High air pressure is associated with fair, stable weather.”
- We make *predictions* (these are logical implications that follow from the truth of the hypothesis)
 - We will find high air pressure on days when the weather is fair and stable
- We go out and test this
 - Go on a fair and stable day, check whether we observe high air

If we get the observation we expect given our hypothesis, we say it has been confirmed!

HYPOTHESES & LAWS



- Another example:
 - We start with a *hypothesis*: “Iron turns red when heated.”
 - We make *predictions*:
 - If we heat an iron rod, it will turn red.
 - We go out and *test* this:
 - We heat up an iron rod, and see if we observe a red color.
 - If we get the observation we expect given our hypothesis, we say it has been *confirmed*!

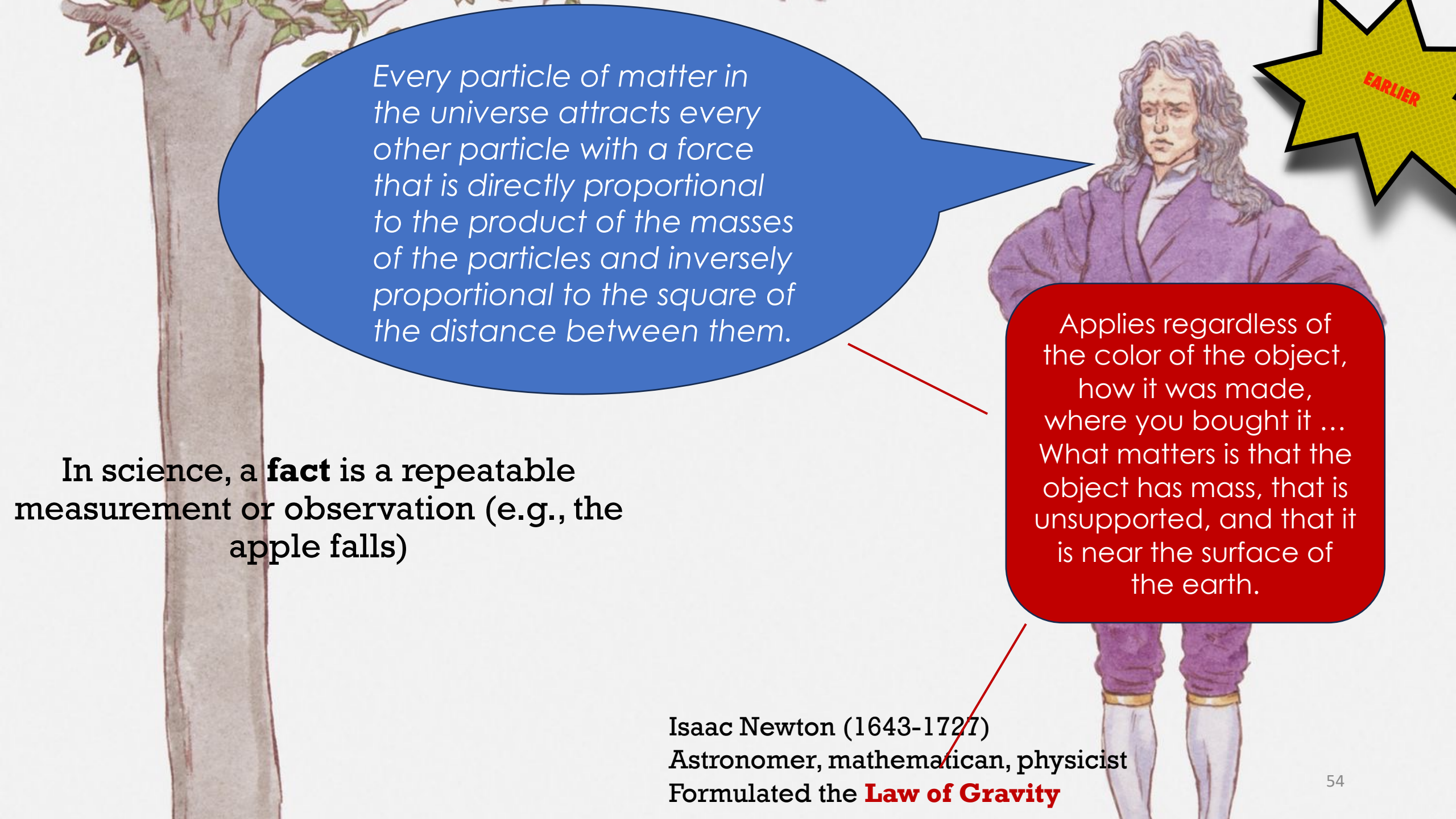
HYPOTHESES & LAWS

- Theories come in degrees of generality.
- Theories, which generalize and describe kinds of things, are **laws**.

HYPOTHESES & LAWS

- **Law (Def.):**

- A claim that associates a general kind of thing with a particular behavior or property is a law.
 - “All metals conduct electricity,” “All plants contain chlorophyll,” “A body’s acceleration varies inversely with its mass.”



Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them.

EARLIER

Applies regardless of the color of the object, how it was made, where you bought it ... What matters is that the object has mass, that is unsupported, and that it is near the surface of the earth.

In science, a **fact** is a repeatable measurement or observation (e.g., the apple falls)

Isaac Newton (1643-1727)
Astronomer, mathematician, physicist
Formulated the **Law of Gravity**

HYPOTHESES & LAWS

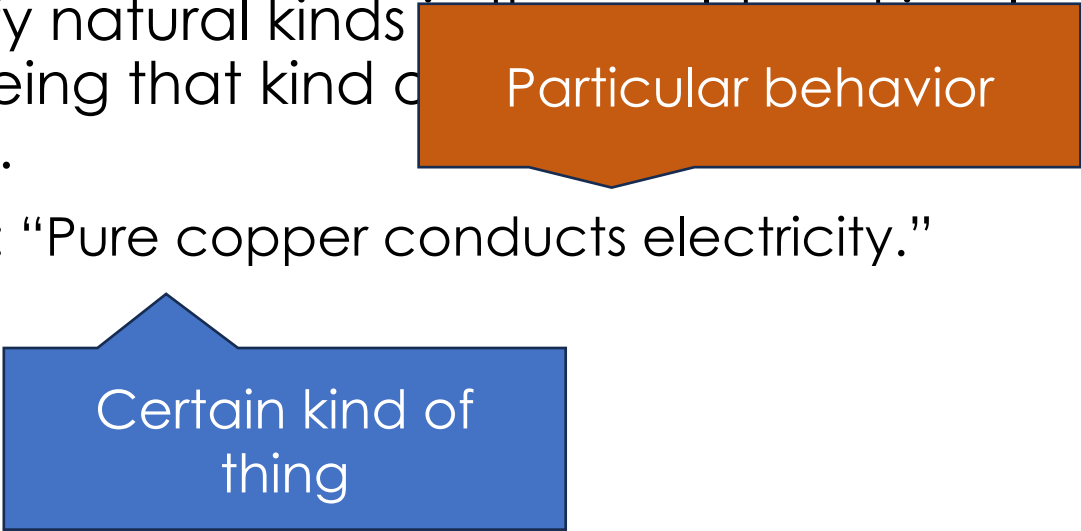
- **Law (Def.):**

- A claim that associates a general kind of thing with a particular behavior or property is a law.
- Laws identify natural kinds in the world and imply a causal connection between being that kind of thing and having the associated property or behavior.
 - Example: "Pure copper conducts electricity."

HYPOTHESES & LAWS

- **Law (Def.):**

- A claim that associates a general kind of thing with a particular behavior or property is a law.
- Laws identify natural kinds that have a causal connection between being that kind of thing and a particular associated property or behavior.
 - Example: "Pure copper conducts electricity."



Particular behavior

The diagram consists of two callout boxes. An orange box with a downward-pointing arrow is positioned to the right of the text 'between being that kind of thing and a particular associated property or behavior'. A blue box with an upward-pointing arrow is positioned below the text 'Certain kind of thing'.

Certain kind of
thing

HYPOTHESES & LAWS

- **Law (Def.):**

- A claim that associates a general kind of thing with a particular behavior or property is a law.
- Laws identify natural kinds in the world and imply a causal connection between being that kind of thing and having the associated property or behavior.
- Claims about accidental associations and the circumstantial properties of such groups are not laws, even though they are generalizations.
 - Example: Everyone in the room is over ten years old.

HYPOTHESES & LAWS

- **Law (Def.):**

- A claim that associates a general kind of thing with a particular behavior or property is a law.
- Laws identify natural kinds in the world and imply a causal connection between being that kind of thing and having the associated property or behavior.
- Claims about accidental associations and the circumstantial properties of such groups are not laws, even though they are generalizations.
 - Example: Everyone in the room is over ten years old.

No causal connection between
being in the room
(circumstantial feature) and
being over ten years old.

HYPOTHESES & LAWS

- **Law (Def.):**

- A claim that associates a property with a behavior or property.
- Laws identify natural connections between being in the room and behavior.
- Claims about accidents of such groups are not laws, even though they are generalizations.
 - Example: Everyone in the room is over ten years old.

It doesn't give us a license to make a prediction like "The next person to come in the room will be over ten years old" since being over ten years old was just a circumstantial feature of the people now in the room.

No causal connection between being in the room (circumstantial feature) and being over ten years old.

HYPOTHESES & LAWS

- **Law (Def.):**

- A claim that associates a general kind of thing with a particular behavior or property is a law.
- Laws identify natural kinds in the world and imply a causal connection between being that kind of thing and having the associated property or behavior.
 - Example: "Pure copper conducts electricity."

The conductivity of copper, on the other hand, does warrant the prediction that the next piece of copper will conduct electricity.

HYPOTHESES & LAWS

- **Law (Def.):**

- A claim that associates a general kind of thing with a particular behavior or property is a law.
- Laws identify natural kinds in the world and imply a causal connection between being that kind of thing and having the associated property or behavior.
- Claims about accidental associations and the circumstantial properties of such groups are not laws, even though they are generalizations.
 - ***Laws support prediction; accidental generalization do not.***

- To sum up:
 - The concepts of theory (a claim about events that are not apparent to experience), hypothesis (not well-tested), and law (a generalization about a natural kind of object) describe **different aspects** of scientific claims and are not exclusive of each other.
 - There is no graduation in terms of ranks between them (first being hypothetical, then being a theory, and finally being a law).
 - The general theory of relativity is an example of a hypothetical law.
 - No theory reaches the level of certainty, of being beyond doubt and immune from change.
 - always a need for the process of justification